

RECOVERY FINE-TUNING RECOVERS WHERE IT WAS TRAINED: DURATION- AND DOMAIN-DEPENDENT GAINS IN PRUNED TEXT-TO-AUDIO DIFFUSION

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ABSTRACT

Structured pruning of text-to-audio diffusion models is followed by recovery fine-tuning, whose benefit is usually certified by one score at one inference setting. Using the released pruned and fine-tuned AudioLDM-M checkpoints at two pruning severities (65% and 83% of U-Net parameters removed), we measure the recovery gain in text-audio alignment with a paired, common-noise protocol across clip duration and prompt domain. The gain is large at the fine-tuning duration of 10.24s (CLAP 0.055 $\rightarrow$ 0.299 at 83% pruning), several-fold smaller at 3.84s (0.015 $\rightarrow$ 0.100), and absent or negative on held-out music. The duration dependence was pre-specified, replicates on a disjoint prompt set, survives a family-wise correction, rank-based statistics and the pruning-seam convention, and is corroborated by a second scorer, event-level metrics and a frame-level grounding model. A matched dense control shows that the fine-tuned checkpoint responds to clip duration at least as strongly as the unpruned model, whereas the 83%-pruned checkpoint barely responds at all. Recovery should be reported across operating points; lacking a matched dense fine-tuned control and human ratings, our claims concern evaluation, not mechanism.

Index Terms— text-to-audio generation, diffusion models, structured pruning, recovery fine-tuning, evaluation

1. INTRODUCTION

Structured pruning followed by recovery fine-tuning is the standard route to cheaper diffusion models [1–3], recently applied to text-to-audio (TTA) generation with AudioLDM [4, 5]. The recovery stage is judged by the restoration of one aggregate number—a Fréchet distance or a KL divergence on the in-domain test set at one sampler setting. A generative model, however, is used at an inference operating point: a clip duration, a prompt domain, a sampler configuration. Nothing guarantees that a recovery gain measured at one point describes the fine-tuned checkpoint at another.

We ask: does the gain of recovery fine-tuning over the pruned checkpoint depend on the operating point at which it is evaluated? We take the released pruned and fine-tuned AudioLDM-M-Full checkpoints of Singh et al. [5] at two pruning severities as the object of study, train nothing, and measure the paired recovery gain in text-audio alignment across two clip durations and two prompt domains. Our findings are:

- **Duration.** The recovery gain is several times larger at the fine-tuning duration (10.24s) than at 3.84s. The interaction was pre-specified, replicates on a disjoint prompt set at the stronger severity, survives a family-wise correction, and holds on scale-free statistics. Against a matched dense control, the 83%-pruned checkpoint has lost most of the dense model’s response to clip duration; the fine-tuned checkpoint responds like the dense model.
- **Domain.** At matched duration the gain is present in-domain (AudioCaps) and absent or negative on held-out music, at both severities and, at severity 2, at both durations. This is not a scoring floor: the pruned checkpoint scores higher on music than in-domain at 10.24s.
- **Where the gain lives.** A frame-level grounding model shows the native-duration gain spread uniformly over the clip, and a crop analysis shows the short-duration deficit arises from generating short clips, not from scoring short excerpts: the first 3.84s of a native generation already carries the whole gain at severity 1 and over half of it at severity 2.

2. BACKGROUND

Pruning and recovery in diffusion models. Structural pruning of diffusion backbones with fine-tuning recovery is established for image models [1–3]. For TTA, Singh et al. [5] L1-prune the AudioLDM U-Net, fine-tune on AudioCaps for 10^6 steps and report in-domain FAD/KL at a fixed sampler; notably their fine-tuned pruned model surpasses the unpruned model in-domain (FAD 1.57 vs. 3.95), a first sign that the recovery stage also acts as further in-domain specialisation. All these works evaluate recovery at one operating point.

Evaluating text-to-audio generation. CLAP similarity [6] is the standard automatic proxy for text-audio alignment; distribution metrics such as FAD [7] are sensitive to sample size and embedding choice [8, 9]; event classifiers [10], human-aligned CLAP [11] and frame-level grounding models [12] measure other axes, motivating a paired, multi-metric design with one pre-specified primary endpoint.

Duration as an operating point. Latent audio diffusion conditions generation on the latent length and is trained at a fixed duration (10.24s for AudioLDM [4, 13]; newer systems condition on timing explicitly [14]), so duration is a mechanistically motivated factor for a checkpoint fine-tuned at one

duration; fine-tuning also trades in-distribution gains for out-of-distribution losses [15], motivating the domain axis.

3. EXPERIMENTAL DESIGN

3.1. Systems

The dense model is AudioLDM-M-Full [4] (U-Net stage multipliers (1, 2, 3, 5), 415.96 M U-Net parameters), itself AudioCaps-fine-tuned by its authors. We study the two released pruning severities of [5]: severity 1, (1, 2, 3, 1), 145.67 M parameters (−65.0%), and severity 2, (1, 2, 1, 1), 71.08 M (−82.9%; the authors report 39% fewer multiply–accumulates at this budget). At each severity we compare the pruned checkpoint (P) with the fine-tuned checkpoint (P+FT), the released result of 10^6 steps of AudioCaps recovery fine-tuning of P [5] (EMA weights, checksums verified). Since the released pruned checkpoints carry no usable EMA weights, P is the released L1 channel selection applied to the dense EMA weights: at severity 1 this reproduces the released raw pruned weights bit-exactly (690/690 tensors); at severity 2 the released checkpoint differs in three decoder-seam tensors, so both conventions are evaluated (A’ primary, B’ sensitivity). “Recovery” names the fine-tuning stage, not an achieved restoration; we train no system.

3.2. Operating points and prompt batteries

Two durations: the native point (10.24s, latent length 256; the fine-tuning duration) and a short point (3.84s, latent length 96). Two domains: in-domain AudioCaps test captions [16] and held-out hip-hop/rap captions from MusicCaps [17], a sub-domain with near-zero exposure in the recovery corpus. Severity 2 uses 192 AudioCaps prompts (disjoint from all severity-1 prompts) and 64 music prompts (3 replicates at 3.84s, one at 10.24s); severity 1 uses 96 AudioCaps prompts for the pre-specified domain test, a matched 80-prompt subset for the duration comparison (on which the dense model is also generated at both durations, giving a matched dense control), and 64 music prompts at 3.84s. Generation uses DDIM [18], 50 steps, guidance 2.5, $\eta = 0$, fp32, single generation, EMA weights (off the published recipe, Sec. 5).

3.3. Paired protocol and scoring

Within each operating point the compared systems receive identical generation noise x_T per prompt (common random numbers), so a per-prompt contrast is not confounded by sampling noise; across durations the latent shapes differ, so duration contrasts are prompt-paired but not noise-paired. The primary scorer is fused CLAP (laion/clap-htsat-fused, rev. 365dea6e) [6], which repeat-pads 3.84s audio to 10s and centre-crops 10.24s audio to 10s; every (system, operating point) is scored with one fixed-order, fixed-seed call, part of the endpoint (re-scoring the same clips in another batch shifts means by ≤ 0.002). The unit is the prompt (music replicates averaged first); intervals are 95% prompt-level percentile bootstrap intervals ($B = 10^4$).

Table 1. Analysis plan (“Pre.”: estimand, gate and prompts committed before any score was seen).

Analysis	Pre.	Role
Sev. 1 domain test (3.84s)	yes	reversal hypothesis; failed
Sev. 1 duration J	yes	follow-up; gate $\log_5 > 0$
Sev. 2 $J, R_{\text{nat}}, R_{\text{short}}, R_{\text{mus}}$, seam B’	yes	primary, disjoint set
Sev. 2 music at 10.24s	yes	design completion; no gate
Human-CLAP, KL, PANNs, FAD	yes	secondary; no gate
FineLAP frame-level profile	yes	post-result diagnostic
Dense duration control; crop; win-rate/rank J ; Holm	no	post-hoc sensitivity

3.4. Quantities and analysis plan

Let R be the mean per-prompt recovery gain, $\text{CLAP}(P+FT) - \text{CLAP}(P)$, at a given (domain, duration); R_{nat} , R_{short} and R_{mus} denote AudioCaps-native, AudioCaps-short and music-short. The duration interaction is

$$J = R_{\text{nat}} - R_{\text{short}} = s(P+FT) - s(P), \quad (1)$$

where $s(\cdot)$ is a system’s duration response, its mean per-prompt score change from 3.84 to 10.24s; J is a paired per-prompt contrast with pre-specified gate $\log_5(J) > 0$. The domain contrast at matched duration is $R_{\text{short}} - R_{\text{mus}}$; the smallest effect size of interest is 0.025. “Pre-specified” means that estimand, gate and prompt set were committed to the version-controlled repository before any score was seen (Table 1).

4. RESULTS

4.1. The recovery gain grows with clip duration

Fig. 1 and Table 2 give the primary result. At severity 2, on the disjoint 192-prompt set with estimands frozen before scoring, the interaction is clearly resolved, $J = +0.159$ [$+0.131, +0.187$]: fine-tuning raises CLAP alignment from 0.055 to 0.299 at the native duration ($R_{\text{nat}} = +0.244$) but only from 0.015 to 0.100 at 3.84s ($R_{\text{short}} = +0.085$). The two duration responses tell the story: the pruned checkpoint gains just $s(P) = +0.040$ [$+0.029, +0.052$] from the longer clip, the fine-tuned one $s(P+FT) = +0.200$ [$+0.173, +0.226$]. The result is seam-robust (B’: $J = +0.161$). The gain is smaller at the short point, not absent (R_{short} resolved; 90% CI [$+0.069, +0.102$] outside ± 0.025).

At the milder severity 1 the same pattern appears with smaller effects: $R_{\text{short}} = +0.008$ [$-0.023, +0.039$], $R_{\text{nat}} = +0.052$ [$+0.009, +0.093$], $J = +0.044$ [$-0.001, +0.087$], whose interval narrowly includes zero. Here the matched dense control anchors the scale. On the same 80 prompts, scored under the same convention, the dense model’s duration response is $s(\text{dense}) = +0.150$ [$+0.100, +0.198$] ($0.202 \rightarrow 0.352$), indistinguishable from the pruned checkpoint’s $s(P) = +0.149$ [$+0.119, +0.178$] (paired difference -0.001 [$-0.056, +0.055$]), while the fine-tuned checkpoint responds more, $s(P+FT) = +0.193$ [$+0.152, +0.232$], exceeding dense by $+0.043$ [$-0.020, +0.109$]. In relative terms, fine-tuning closes 8% [$-30\%, +36\%$] of the pruned checkpoint’s gap to dense at 3.84s and 52% [$+11\%, +103\%$] at 10.24s. At severity 2, $s(P) = +0.040$ is far below the dense response measured on the severity-1 prompts, while

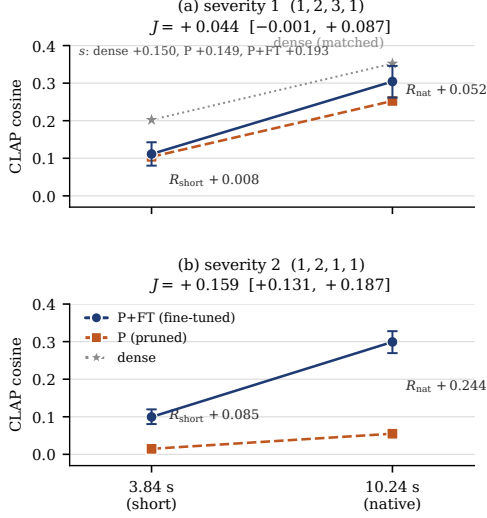


Fig. 1. The recovery gain grows with clip duration. Mean CLAP cosine of the pruned (P, dashed) and fine-tuned (P+FT, solid) checkpoints at the short and native points for (a) severity 1 and (b) severity 2; whiskers: 95% CI of the paired gain R , drawn about the P+FT mean; J annotated. Panel (a) includes the matched dense control (grey; same 80 prompts, same scoring convention): its duration response equals the pruned checkpoint’s and is below the fine-tuned one’s.

$s(P+FT) = +0.200$ is of dense magnitude (a cross-set, descriptive comparison).

Because every system scores higher at 10.24s, a raw-scale interaction could reflect scale alone, so we report it on scale-free statistics as well (post-hoc). The fraction W of prompts on which P+FT beats P rises from 0.44 to 0.64 at severity 1 ($\Delta W = +0.20 [+0.06, +0.34]$) and from 0.72 to 0.87 at severity 2 ($+0.15 [+0.07, +0.22]$); the median per-prompt interaction is $+0.051 [+0.012, +0.077]$ and $+0.172 [+0.153, +0.214]$ (Wilcoxon $p = 0.02$ and $p < 10^{-17}$); on pooled-rank scores J is $+0.063 [-0.015, +0.143]$ and $+0.199 [+0.153, +0.244]$; the second scorer gives $J_{\text{HC}} = +0.075 [+0.012, +0.137]$ at severity 1. A Holm correction over all 13 contrasts in this paper leaves every severity-2 conclusion unchanged (bootstrap $p < 10^{-4}$); at severity 1, R_{nat} ($p = 0.016$) and J ($p = 0.052$) do not survive it. The severity-1 duration effect is therefore directionally consistent on every scale but is established by the severity-2 replication.

4.2. At matched duration the gain is domain-specific

Held-out music separates the domains at fixed duration. At severity 2 and 3.84s the gain is resolved in-domain ($R_{\text{short}} = +0.085$) and null on music ($R_{\text{mus}} = +0.009 [-0.013, +0.032]$; $W = 0.48$), a matched-duration domain contrast of $+0.076 [+0.047, +0.105]$. At severity 1 the pre-specified domain test (96 prompts) found no in-domain gain at 3.84s ($R_{\text{AC}} = -0.002 [-0.027, +0.021]$) and a negative music contrast ($R_{\text{mus}} = -0.094 [-0.124, -0.065]$), a domain contrast of $+0.092 [+0.054, +0.131]$. The music penalty is severity-specific; the absence of a music gain replicates. At the native duration the music contrast is likewise null

Table 2. Recovery gain in CLAP alignment: mean cosine per checkpoint, paired gain $R = \text{CLAP}(P+FT) - \text{CLAP}(P)$ with 95% CI, and fraction W of prompts on which P+FT beats P. Duration-response rows give $s(P)$, $s(P+FT)$ and, in the R column, J (matched dense response at severity 1: $+0.150$). †: CI excludes 0; ‡: pre-specified gate passes. Under a Holm correction over all 13 contrasts, every severity-2 † survives and at severity 1 only the music rows do. J paired per prompt; domain contrasts two-sample. n : sev. 1 AudioCaps 80 (96 for the domain contrast), sev. 2 AudioCaps 192, music 64.

Setting	P	P+FT	R [95% CI]	W
Severity 1, (1, 2, 3, 1)				
AudioCaps 3.84s	0.104	0.111	$+0.008 [-0.023, +0.039]$	0.44
AudioCaps 10.24s	0.253	0.304	$+0.052^\dagger [+0.009, +0.093]$	0.64
music 3.84s	0.117	0.023	$-0.094^\dagger [-0.124, -0.065]$	0.20
duration s ; J	$+0.149$	$+0.193$	$+0.044 [-0.001, +0.087]$	
domain 3.84s ($n=96$)			$+0.092^\dagger [+0.054, +0.131]$	
Severity 2, (1, 2, 1, 1)				
AudioCaps 3.84s	0.015	0.100	$+0.085^\dagger [+0.066, +0.105]$	0.72
AudioCaps 10.24s	0.055	0.299	$+0.244^\dagger [+0.215, +0.273]$	0.87
music 3.84s	0.005	0.014	$+0.009 [-0.013, +0.032]$	0.48
music 10.24s	0.089	0.094	$+0.005 [-0.028, +0.039]$	0.53
duration s ; J	$+0.040$	$+0.200$	$+0.159^\dagger [+0.131, +0.187]$	
domain 3.84s			$+0.076^\dagger [+0.047, +0.105]$	
domain 10.24s			$+0.239^\dagger [+0.195, +0.283]$	

($+0.005 [-0.028, +0.039]$; $W = 0.53$), so the large native gain is confined to the fine-tuning domain: the 10.24s domain contrast is $+0.239 [+0.195, +0.283]$, and the music duration interaction is $-0.004 [-0.043, +0.034]$. Two artefacts are excluded. The music null is not a scoring floor: at 10.24s the pruned checkpoint scores higher on music (0.089) than in-domain (0.055), and music scores have full spread. Nor is it a caption-length effect: music captions are seven times longer than AudioCaps captions (median 56.5 vs. 8 words), but within AudioCaps the per-prompt native gain is uncorrelated with caption length (Spearman $\rho = +0.04 [-0.12, +0.18]$, severity 2); “domain” nevertheless bundles content and caption style.

4.3. Where in the clip the gain lives

Two post-result analyses locate the native-duration gain. First, as a prospectively frozen diagnostic, we score the native clips with FineLAP [12], a frame-level language–audio grounding model (EAT encoder, not CLAP-derived) that reports per 0.16s frame how strongly the requested event is grounded (110 and 49 eligible prompts at severities 2 and 1, selected outcome-blind). Fine-tuning raises grounding across the whole clip (severity 2: semantic mass $+0.27 [+0.22, +0.33]$, occupancy $+0.27$, quarter coverage $+0.38$, peak $+0.41$; CIs exclude 0; both severities; seam-robust), and the gain in the first 3.84s equals the gain in the remainder ($D_{\text{early}} = +0.275$ vs. $D_{\text{late}} = +0.273$; $T = -0.002 [-0.024, +0.020]$, Fig. 2a); the native gain is not confined to the extra seconds. Second, a post-hoc crop analysis separates generation length from scoring window: we score only the first 3.84s of each 10.24s generation with the same CLAP convention and compare with the separately generated 3.84s clips of the same prompts (Fig. 2b). The crop carries a large part of the native gain. At severity 1, $R_{\text{crop}} = +0.051 [+0.016, +0.085]$ equals R_{nat}

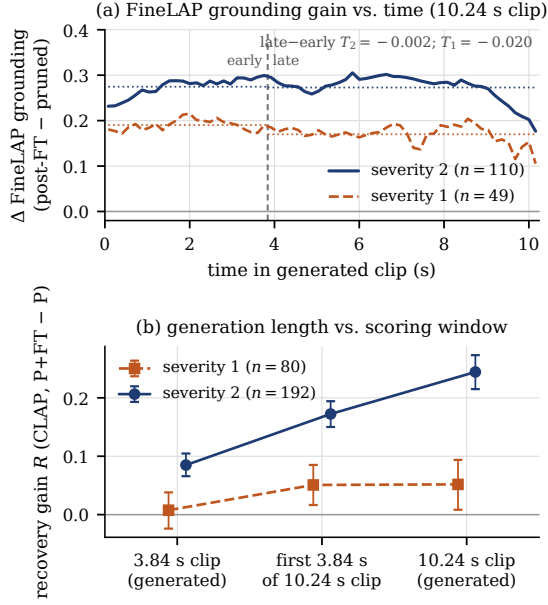


Fig. 2. (a) FineLAP frame-level grounding gain (P+FT - P) vs. time in the 10.24 s clip (dotted: window means; vertical line: 3.84 s): uniform, not back-loaded. (b) Recovery gain R (95% CI) on the generated 3.84 s clip, on the first 3.84 s of the 10.24 s clip, and on the full 10.24 s clip: the crop carries the whole gain at severity 1 and over half at severity 2—a generation-length, not a scoring-window, effect.

($R_{\text{nat}} - R_{\text{crop}} = +0.001 [-0.027, +0.029]$) and exceeds R_{short} by $+0.043 [+0.002, +0.085]$; at severity 2, $R_{\text{crop}} = +0.172 [+0.150, +0.194]$, with $R_{\text{crop}} - R_{\text{short}} = +0.087 [+0.065, +0.110]$ and $R_{\text{nat}} - R_{\text{crop}} = +0.072 [+0.055, +0.089]$ (seam-robust). The short-duration deficit is thus mostly a property of what the fine-tuned checkpoint generates when asked for a short clip, not of scoring a short excerpt—consistent with a checkpoint specialised to its fine-tuning duration.

4.4. Corroboration and pre-specified negatives

CLAP is the sole primary endpoint; secondaries corroborate without gating. At the severity-2 native point Human-CLAP [11], a CLAP model fine-tuned on human ratings (automatic, not a listening test), gives $R_{\text{nat}} = +0.375 [+0.340, +0.408]$ and $J = +0.185$; two pre-specified event-level metrics favour P+FT with paired CIs excluding zero—KL to real references 2.23 vs. 4.45 ($\Delta = +2.22 [+1.93, +2.53]$) and PANNs top-10 captured labels 1.46 vs. 0.60 ($\Delta = +0.86 [+0.70, +1.02]$); FAD agrees (6.92 vs. 27.4, descriptive).

Three pre-specified hypotheses failed and are kept. (i) That recovery fine-tuning trades in-domain for out-of-domain alignment: it requires an in-domain gain at 3.84 s, and severity 1 has none ($R_{\text{AC}} \approx 0$, Sec. 4.2); the loss on music is real, the trade is not. (ii) The severity-1 music penalty did not replicate (Human-CLAP alone shows a small negative music contrast at severity 2, $-0.037 [-0.068, -0.005]$; the sign is unestablished). (iii) A “late allocation” account of

the duration effect is rejected by FineLAP ($T \approx 0$). Finally, P+FT is not restored to dense: at severity 1 and 10.24 s the dense model leads P by $+0.099 [+0.058, +0.140]$ and P+FT by $+0.048 [-0.000, +0.096]$ —neither a resolved gap nor equivalence.

5. DISCUSSION AND LIMITATIONS

Across severities, scorers and metrics, the recovery gain of the released fine-tuned AudioLDM-M checkpoints tracks the operating point of the fine-tuning itself: it is largest at the fine-tuning duration, confined to the fine-tuning domain, and heterogeneous across prompts (column W). The dense control sharpens the reading. At 65% pruning the pruned checkpoint still responds to clip duration exactly like the dense model, and fine-tuning adds a modest, in-domain-only increment. At 83% pruning the pruned checkpoint has lost most of that response, and fine-tuning restores a response of dense magnitude—again only in the fine-tuning domain. This is consistent with the recovery stage acting, at least in part, as further in-domain, native-duration specialisation. Post-pruning recovery is thus better described as multi-operating-point behaviour than as a single restored score. We recommend that recovery studies report (1) the training-matched operating point and one motivated off-native point (another duration or domain); (2) absolute scores of P, P+FT and, where available, the dense model at each point; (3) paired gains with prompt-level uncertainty under common random numbers; and (4) an explicit interaction across points—an implication of one case study (one model family, two severities, two durations, two domains).

Limitations. Mechanistic attribution is blocked: the matched control—the dense model given the same AudioCaps fine-tune—was trained by the original authors but no longer exists, and the public dense text-fine-tuned release is not equivalent (different data and recipe), so the dependence cannot be attributed to pruning rather than to fine-tuning in general. The dense duration control exists only on the severity-1 prompt set; its comparison with the severity-2 responses is cross-set. Primary inference rests on CLAP-family scorers (FineLAP is the only non-CLAP model) and there is no human evaluation. Sampler settings differ from the published recipe (DDIM 50 vs. 200, guidance 2.5 vs. 3.5, single vs. best-of-3); both systems are always evaluated identically. Only two durations were tested; cross-severity comparisons are descriptive (different prompt sets); the severity-1 duration effect does not survive family-wise correction; the frame-level, crop and dense-control analyses are post-result.

6. CONCLUSION

For the released pruned and fine-tuned AudioLDM-M checkpoints, recovery fine-tuning recovers much at its own operating point and much less away from it: the CLAP gain is several times larger at the fine-tuning duration than at 3.84 s and absent or negative on held-out music; at 83% pruning it restores, in-domain only, a dense-like response to clip duration. A single in-domain, native-duration score overstates what the fine-tuned checkpoint delivers elsewhere. Audio examples: <https://gbibbo.github.io/audioldm-modality-swap-pruning/>.

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